

2. (a)

Taking the two particles as A and B, according to **Newton's 3<sup>rd</sup> law**, when the two particles collide, the force that A exerts on B is equal in magnitude but opposite in direction to the force that B exerts on A.

According to **Newton's 2<sup>nd</sup> law**, the net/resultant force on a particle is proportional to the rate of change of its momentum.

Since the **time of contact for A and B is equal**, the change in momentum of A is exactly equal in magnitude, and opposite in sign/direction, to the change in momentum of B.

Hence the total momentum is conserved.

(b) (i) **Total kinetic energy** of the **system** is not conserved in the collision. [1]

(ii) By the principle of conservation of momentum,

$$M_1 u = (M_1 + M_2) v$$

$$(7.0 \times 10^5)(3.0) = (7.0 \times 10^5 + M_2)(2.0)$$

$$M_2 = 3.5 \times 10^5 \text{ kg.}$$

(iii) Drawing of a suitable tangent at  $t = 1.0 \text{ s}$  and showing of relevant working.  
(Tangent triangle should be as large as possible and coordinates read to half a small square).

Magnitude of acceleration =  $0.65 \text{ m s}^{-2}$

Direction: left

(iv) Magnitude of retarding force =  $7.0 \times 10^5 \times 0.65 = 4.6 \times 10^5 \text{ N}$